

OPSPILLOT AI

FRESH SETUP COMPLETE GUIDE - ENHANCED AI REMEDiation V4

*AIOPS FOR DEVOPS: PROJECT HEALTH GUARDIAN, AI RCA, HUMAN-APPROVED
REMEDiation, DOCKER, GITHUB ACTIONS, AWS EC2*

Audience: This guide is for a fresh project/repository setup using the provided source ZIP. It is written for students and trainers who need spoon-feeding steps and troubleshooting clarity.

Core idea: When the main pipeline fails, OpsPilot AI should create RCA, wait for /ai-agent approve, fix allow-listed repository issues, validate, and open a PR. Humans still merge and approve deployments.

Security: Never commit API keys, Docker tokens, PATs, SSH private keys, or passwords. Use GitHub Secrets and protected Environments.



TABLE OF CONTENTS

1. Final project outcome
2. Source ZIP contents
3. Tech stack required
4. Required accounts, tokens, and secrets
5. AWS EC2 preparation
6. Extract source and initialize Git
7. Local Python setup and app test
8. Local pytest and project doctor
9. Local Docker build/run test
10. Push source to GitHub
11. GitHub Actions pipeline and approvals
12. Enhanced AI Remediation v4 architecture
13. Project Health Guardian
14. AI RCA and human-approved remediation flow
15. Controlled failure tests
16. Troubleshooting matrix
17. What AI can and cannot fix
18. Final checklist and submission evidence

1. FINAL PROJECT OUTCOME

- A new GitHub repository containing the complete OpsPilot AI source code.
- A Flask app running locally, in Docker, and on AWS EC2.
- A CI/CD workflow with validate, pytest, Docker publish, and production deploy jobs.
- Human approval gates for Docker publishing and production deployment.
- AI RCA workflow that creates an issue/artifact after failures.
- AI remediation workflow that fixes allow-listed project files after /ai-agent approve.
- Project Health Guardian workflow that detects broken YAML/policy/config issues.
- AI Artifact Generator practice: students can ask the app to draft Dockerfiles/YAML and validate them safely.

High-level flow:

Push to main



Project Health Guardian checks repository policy/YAML



Main pipeline validate + pytest



Human approves docker-publish



Docker image pushed to Docker Hub



Human approves production



EC2 deploy and /health verification

If failure occurs:



AI RCA issue/artifact



Human comments /ai-agent approve



AI remediation validates and opens PR



Human reviews and merges PR

2. SOURCE ZIP CONTENTS

- Files included in OpsPilot_AI_v4_Fresh_Source.zip

Item	Details
app.py	Flask app with Gemini integration and /health endpoint.
Dockerfile	Production container build using gunicorn, non-root

	user, EXPOSE, and HEALTHCHECK.
requirements.txt	Runtime dependencies: Flask, google-generativeai, gunicorn.
requirements-dev.txt	Developer/CI dependencies: runtime requirements, pytest, PyYAML.
tests/test_app.py	Pytest tests for /health, home page, and missing key behavior.
.github/workflows/deploy.yml	Main CI/CD pipeline: validate + pytest + Docker publish + EC2 deploy.
.github/workflows/ai-agent-rca.yml	Creates RCA artifact/issue when the main pipeline fails.
.github/workflows/ai-agent-remediate.yml	Human-approved remediation workflow with validation retry loop.
.github/workflows/ai-agent-guardian.yml	Project Health Guardian workflow for YAML/policy/project checks.
scripts/ai_ci_agent.py	AI/Ops engine: analyze, remediate, doctor modes.
docs/AI_AGENT_RUNBOOK.md	Operations runbook for maintainers and trainers.
.gitignore / .dockerignore	Keeps runtime artifacts, caches, and secrets out of Git/Docker context.



OpsPilotAI.zip

3. TECH STALK REQUIRED

Requirements needed :-

Item	Details
GitHub	Source, Actions, issues, PRs, secrets, environments.
Docker Hub	Container registry for published images.
Cloud (AWS or Azure)	EC2 or VM Ubuntu server.
Google AI Studio / Gemini	Gemini API key for app and AI agent.
Python Language installed	Application based on python Language

```
D:\aiops>docker --version
Docker version 29.6.2, build dfc4efb
D:\aiops>
```

- Go to <https://github.com/new>.
- Repository name suggestion: OpsPilot-AI or opspilot-ai.
- Choose Public or Private as your lab requires.
- Do not initialize with README, .gitignore, or license. The source ZIP already contains these files.
- Create the repository and copy the HTTPS remote URL.

Repo should be empty: If GitHub creates README/.gitignore for you, remove them or create a different empty repo to avoid conflicts.

4. VARIABLES, TOKENS, AND SECRETS

GitHub Variables:-

Variable	Recommended value	Purpose
AI_AGENT_MODEL	AI model	Model used by RCA/remediation. Avoid older unavailable model names.
GEMINI_MODEL	AI model	Model used by the deployed Flask app.
AI_AGENT_MAX_ATTEMPTS	3	Maximum remediation retry attempts.

AI Models for AI agent Model and gemini model :-

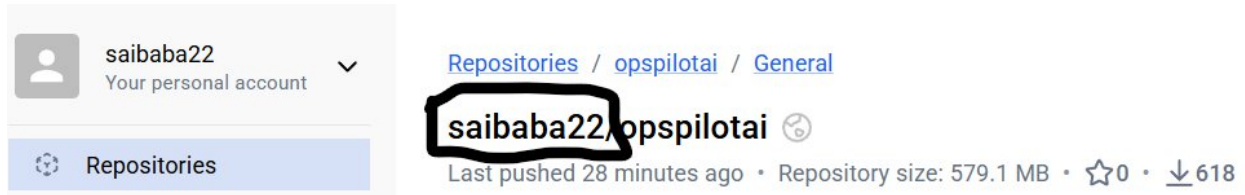
- Ⓟ Flash & Lite Models (Standard Free Tier)
 - Gemini 2.0 Flash / 2.0 Flash 001
 - Gemini 2.0 Flash-Lite / 2.0 Flash-Lite 001
 - Gemini 2.5 Flash / 2.5 Flash-Lite
 - Gemini 3.1 Flash-Lite / 3.1 Flash-Lite Preview
 - Gemini 3.5 Flash / 3.5 Flash-Lite
 - Gemini 3.6 Flash
 - Gemini Flash Latest / Gemini Flash-Lite Latest
- Ⓟ Pro & Preview Models (Currently Free in AI Studio Preview)
 - Gemini 2.5 Pro
 - Gemini 3 Pro Preview
 - Gemini 3.1 Pro Preview
 - Gemini Pro Latest
- Gemini Omni Flash Preview
- Deep Research Preview / Deep Research Pro Preview
- Ⓟ Open Weights Models (Free to use)
 - Gemma 4 26B A4B IT
 - Gemma 4 31B IT
- Ⓟ Specialized Models
 - Gemini Embedding 001 / 2 / 2 Preview
 - AQA (Attributed Question Answering)
 - Gemini 2.5 Computer Use Preview
 - Lyria 3 Clip/Pro Preview (Music/Audio)
 - Imagen 4 / 4 Ultra / 4 Fast (Image generation - often has a free daily limit)

GitHub Secrets:-

Secret	Required	Purpose
DOCKERHUB_USERNAME	Yes	Docker Hub username.
DOCKERHUB_TOKEN	Yes	Docker Hub access token, not account password.
GEMINI_API_KEY	Yes	Gemini key used by the deployed Flask app.
AI_AGENT_API_KEY	Recommended	Separate Gemini key for RCA/remediation. If omitted, agent falls back to GEMINI_API_KEY.
AI_AGENT_GITHUB_TOKEN	Yes for remediation changing workflows	PAT/fine-grained token with repo + workflow permission so AI PRs can update .github/workflows/*.yaml.
EC2_HOST	Yes	EC2 public IP or public DNS.
EC2_SSH_KEY	Yes	Full private key content for SSH to EC2.
EC2_USER	Yes	EC2 OS Username
VM_HOST	Yes	VM public ip or public dns

VM_KEY	Yes	VM Password or pem key content
VM_USER	Yes	Your vm username

- EC2_HOST : AWS EC2 IP ADDRESS
- EC2_SSH_KEY : YOUR EC2 PEM KEY
- EC2_USER : for ubuntu – ubuntu , for amazon linux : ec2-user
- VM_HOST : AZURE VM IP ADDRESS
- VM_KEY : VM PASSWORD OR KEY
- VM_USER : AZURE VM USERNAME
- DOCKERHUB_USERNAME : YOUR DOCKER HUB ACCOUNT NAME (This is sample)



- DOCKERHUB TOKEN : Follow instructions to get this
 - Open your Docker hub account, or if you don't have account create it
 - Click on profile pic or profile icon at top right
 - Click on account settings
 - Click on Personal Access Token (PAT)
 - Click on generate new Token
 - Access Token Description : any
 - Expiry date : Your choice
 - Access Permissions : Read and write

Access token description

ai

Expiration date

None

Optional

Access permissions

Read & Write

Read & Write tokens allow you to push images to any repository managed by your account.

Cancel Generate

- Click on Generate button
- Now you will get PAT copy it and note a side (Once token gone or vanishes will not appear again create one more)

Access token description
ai

Expires on
Never

Access permissions
Read & Write

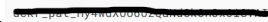
To use the access token from your Docker CLI client:

1. Run

```
$ docker login -u 
```

Copy

2. At the password prompt, enter the personal access token.

```

```

Copy

[Back to access tokens](#)

- **AI_AGENT_GITHUB_TOKEN** : to get this github token follow this instructions
 - Open your Github account
 - Click on profile pic or profile icon at top right
 - Click on settings, From left bar, click on developer settings
 - Now click on PAT - Personal Access Token -> Tokens (Classic)
 - Generate new token -> Classic
 - Name : any
 - Expiry : your choice
 - Select all the options and click on generate.
 - Copy github token starts with ghp
- **GEMINI_API_KEY** and **AI_AGENT_API_KEY** : both are same key
 - Open this page <https://aistudio.google.com/api-keys>
 - If having account, create a project and key
 - If don't have account create one account and create key.
 - Create one project, under it create one api key
 - API key starts from AQ.Ab

Model issue we faced: RCA was generic when the model gemini-1.5-flash was unavailable. Set **AI_AGENT_MODEL=gemini-2.0-flash** to avoid fallback-only RCA.

5. VIRTUAL MACHINE OR SERVER PREPARATION

- This preparation involves 2 clouds (**Note : You can use both clouds or any one cloud its your choice**)
 - A. AWS EC2 instance
 - B. AZURE VM Machine

A. AWS - EC2 instance	B. AZURE - VM Machine
Launch Ubuntu 22.04/24.04 EC2 instance.	Ubuntu 22.04/24.04
Allow SSH port 22 from your IP. Allow TCP port 5000 from your IP for lab testing.	Allow SSH port 22 from your IP. Allow TCP port 5000 from your IP for lab testing.
Download key pair .pem file.	Download key pair .pem file. Or password
Copy EC2 public IP	Copy VM public IP

- ⑨ Set key permissions and SSH from local PC

Bash / Git Bash / Linux / macOS	Windows PowerShell	Windows CMD
chmod 400 <KEY_FILE>.pem ssh -i <KEY_FILE>.pem ubuntu@<EC2_HOST>	icacls .\<KEY_FILE>.pem /inheritance:r icacls .\<KEY_FILE>.pem /grant:r "\$(\$env:USERNAME):(R)" ssh -i .\<KEY_FILE>.pem ubuntu@<EC2_HOST>	icacls <KEY_FILE>.pem /inheritance:r icacls <KEY_FILE>.pem /grant:r "%USERNAME%:R" ssh -i <KEY_FILE>.pem ubuntu@<EC2_HOST>

- ⑨ For AWS EC2 and Azure VM instance configuration inside server

Purpose \ Cloud	AWS EC2	AZURE VM WITH PEM KEY	AZURE VM WITH PASSWORD
For updating :	sudo apt-get update	sudo apt-get update	sudo apt-get update
For Install Docker	sudo apt-get install -y docker.io	sudo apt-get install -y docker.io	sudo apt-get install -y docker.io
To start Docker	sudo systemctl start docker	sudo systemctl start docker	sudo systemctl start docker
Enable Docker	sudo systemctl enable --now docker	sudo systemctl enable --now docker	sudo systemctl enable -- now docker
Create Docker Group	sudo usermod -aG docker ubuntu	sudo usermod -aG docker VM-USERNAME	sudo usermod -aG docker VM-USERNAME
Docker Socket Permission	sudo chmod 666 /var/run/docker.sock	sudo chmod 666 /var/run/docker.sock	sudo chmod 666 /var/run/docker.sock
Check docker containers	sudo docker ps	sudo docker ps	sudo docker ps

- If you are using Azure VM with Password use these commands

- Enable password authentication

```
sudo sed -i 's/^KbdInteractiveAuthentication no/KbdInteractiveAuthentication yes/' /etc/ssh/sshd_config
```

```
sudo sed -i 's/^#KbdInteractiveAuthentication.* /KbdInteractiveAuthentication yes/' /etc/ssh/sshd_config
```

- Restart ssh service

```
sudo systemctl restart ssh
```

- Optional install supporting software :

```
sudo apt install -y ca-certificates curl gnupg lsb-release apt-transport-https software-properties-common
```

- Common for both : 4 or 8GB for better run

6. EXTRACT SOURCE AND INITIALIZE GIT

- Extract the source ZIP

Bash / Git Bash / Linux / macOS	Windows PowerShell	Windows CMD
unzip OpsPilot_AI_v4_Fresh_Source.zip cd OpsPilot-AI	Expand-Archive - Path .\OpsPilot_AI_v4_Fresh_Sourc e.zip -DestinationPath . -Force cd OpsPilot-AI	powershell -Command "Expand- Archive - Path .\OpsPilot_AI_v4_Fresh_Sourc e.zip -DestinationPath . -Force" cd OpsPilot-AI

- Initialize and connect to your new GitHub repo

Bash / Git Bash / Linux / macOS	Windows PowerShell	Windows CMD
git init git branch -M main git remote add origin https://github.com/<YOUR_USER>/ OpsPilot-AI.git git status	git init git branch -M main git remote add origin https://github.com/<YOUR_USER>/ OpsPilot-AI.git git status	git init git branch -M main git remote add origin https://github.com/<YOUR_USER>/ OpsPilot-AI.git git status

7. LOCAL PYTHON SETUP AND APP TEST

- Create virtual environment in project repo or folder.

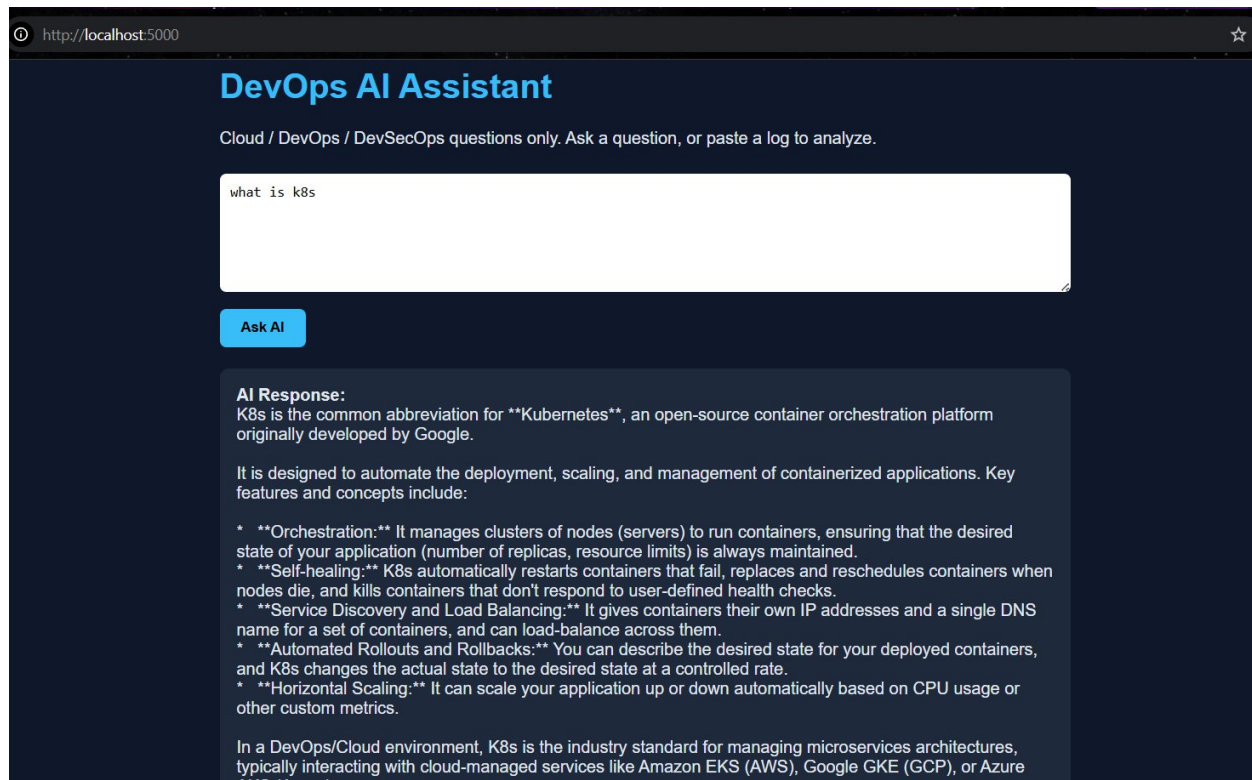
Bash / Git Bash / Linux / macOS	Windows PowerShell	Windows CMD
python3 -m venv venv source venv/bin/activate python -m pip install --upgrade pip	py -3.11 -m venv venv .\venv\Scripts\Activate.ps1 python -m pip install --upgrade pip	py -3.11 -m venv venv venv\Scripts\activate.bat python -m pip install --upgrade pip

Pip install -r requirements.txt
pip install -r requirements-dev.txt

- Install and run app :

Bash / Git Bash / Linux / macOS	Windows PowerShell	Windows CMD
pip install -r requirements.txt pip install -r requirements-dev.txt export GEMINI_API_KEY="<GEMINI_API_KEY>" export GEMINI_MODEL="gemini-2.0-flash" python app.py	pip install -r requirements.txt pip install -r requirements-dev.txt \$env:GEMINI_API_KEY="<GEMINI_API_KEY>" \$env:GEMINI_MODEL="gemini-2.0-flash" python app.py	pip install -r requirements.txt pip install -r requirements-dev.txt set GEMINI_API_KEY=<GEMINI_API_KEY> set GEMINI_MODEL=gemini-2.0-flash python app.py

- Now browse in web as localhost:5000 or 127.0.0.1:5000



- Health check in another terminal

Bash / Git Bash / Linux / macOS	Windows PowerShell	Windows CMD
curl -fsS http://localhost:5000/health	Invoke-RestMethod http://localhost:5000/health	curl -f http://localhost:5000/health

- Let app.py be running
- Open one more cmd and type : `curl -f http://localhost:5000/health`

```
C:\Users\HOME>curl -f http://localhost:5000/health
{"status": "ok"}

C:\Users\HOME>
```

8. LOCAL PYTEST AND PROJECT DOCTOR

- Run tests and project doctor

Bash / Git Bash / Linux / macOS	Windows PowerShell	Windows CMD
pip install -r requirements-dev.txt	pip install -r requirements-dev.txt	pip install -r requirements-dev.txt

pytest -q python scripts\ai_ci_agent.py doctor --output-dir ai-agent- output	pytest -q python scripts\ai_ci_agent.py doctor --output-dir ai-agent- output	pytest -q python scripts\ai_ci_agent.py doctor --output-dir ai-agent- output
---	---	---

pytest -q
Expected:
pytest: 3 passed
Project Health Doctor: Status: PASS

```
(venv) D:\aiops>pytest -q
...
3 passed in 1.85s
```

- Testing again :
python scripts\ai_ci_agent.py doctor --output-dir ai-agent-output

```
(venv) D:\aiops>python scripts\ai_ci_agent.py doctor --output-dir ai-agent-output
# OpsPilot AI Project Health Report

Generated: 2026-07-31T10:49:39Z
Status: PASS
Discovered app/container port: 5000

## Errors
- None

## Warnings
- None

## Info
- Discovered app/container port: 5000
- Port evidence: Dockerfile EXPOSE=5000(w=5); Dockerfile bind/health=5000(w=5); Dockerfile bind/health=5000(w=5); app.py app.run=5000(w=5); deploy.yml health check=5000(w=1); deploy.yml health check=5000(w=1); deploy.yml docker container port=5000(w=2); deploy.yml docker host port=5000(w=1); deploy.yml docker container port=5000(w=2); deploy.yml docker host port=5000(w=1)

## Human approval
If this report was attached to an RCA/Guardian issue and you want the AI agent to prepare a fix, comment exactly:
```text
/ai-agent approve
```
```

- Remove generated doctor output before committing

| Bash / Git Bash / Linux / macOS | Windows PowerShell | Windows CMD |
|---------------------------------|---|-----------------------------|
| rm -rf ai-agent-output | Remove-Item -Recurse -Force ai-agent-output -ErrorAction SilentlyContinue | rmdir /S /Q ai-agent-output |

rmdir /S /Q ai-agent-output

```
(venv) D:\aiops>rmdir /S /Q ai-agent-output
(venv) D:\aiops>|
```

9. LOCAL DOCKER BUILD/RUN TEST

- Build and run Docker locally

| Bash / Git Bash / Linux / macOS | Windows PowerShell | Windows CMD |
|--|--|--|
| <pre>docker build -t opspilot-ai:local . docker run -d -p 5000:5000 -e GEMINI_API_KEY="<GEMINI_API_K EY>" -e GEMINI_MODEL="gemini- 2.0-flash" --name opspilot-local opspilot-ai:local curl -fsS http://localhost:5000/health docker rm -f opspilot-local</pre> | <pre>docker build -t opspilot-ai:local . docker run -d -p 5000:5000 -e GEMINI_API_KEY="<GEMINI_API_K EY>" -e GEMINI_MODEL="gemini- 2.0-flash" --name opspilot-local opspilot-ai:local Invoke-RestMethod http://localhost:5000/health docker rm -f opspilot-local</pre> | <pre>docker build -t opspilot-ai:local . docker run -d -p 5000:5000 -e GEMINI_API_KEY=<GEMINI_API_K EY> -e GEMINI_MODEL=gemini- 2.0-flash --name opspilot-local opspilot-ai:local curl -f http://localhost:5000/health docker rm -f opspilot-local</pre> |

docker build -t opspilot-ai:local .

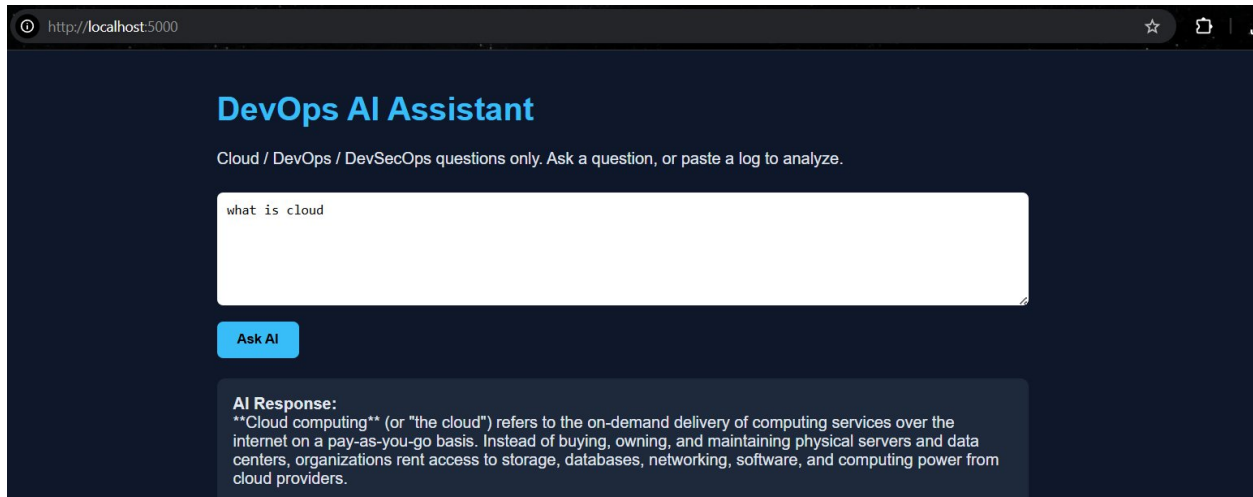
```
D:\aiops>docker build -t opspilot-ai:local .
[+] Building 54.7s (11/12)                                docker:desktop-linux
=> [internal] load build definition from Dockerfile      0.1s
=> => transferring dockerfile: 718B                     0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim 8.2s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore                        0.1s
=> => transferring context: 353B                         0.0s
=> [1/6] FROM docker.io/library/python:3.11-slim@sha256:db3ff2e1800a8581e2c48a27c3995339d47bdf046da21c7627accd3 10.2s
=> => resolve docker.io/library/python:3.11-slim@sha256:db3ff2e1800a8581e2c48a27c3995339d47bdf046da21c7627accd3d 0.1s
=> => sha256:c89b9f64c028c19ba92e195d7589d914b9cd1fc69f3c3dfeb931e93876ac2064 248B / 248B 0.3s
=> => sha256:6b265b8eae4a26a263eb55546aa430c1eef37032a7cd86393d98c6b779f28319 14.42MB / 14.42MB 2.0s
=> => sha256:9775d166087aba0afe5ba9a88859c04c08090e82ae218742d09fec9119e0335d 1.29MB / 1.29MB 1.3s
=> => sha256:062e450697faa5f02a3a74eba9864ee4d79bc9cfbd65769fc6cdf2c05c6a053 29.78MB / 29.78MB 4.7s
=> => extracting sha256:062e450697faa5f02a3a74eba9864ee4d79bc9cfbd65769fc6cdf2c05c6a053 2.7s
=> => extracting sha256:9775d166087aba0afe5ba9a88859c04c08090e82ae218742d09fec9119e0335d 0.4s
=> => extracting sha256:6b265b8eae4a26a263eb55546aa430c1eef37032a7cd86393d98c6b779f28319 2.0s
=> => extracting sha256:c89b9f64c028c19ba92e195d7589d914b9cd1fc69f3c3dfeb931e93876ac2064 0.0s
=> [internal] load build context                        0.2s
=> => transferring context: 4.17kB                       0.1s
=> [2/6] WORKDIR /app                                0.5s
=> [3/6] COPY requirements.txt .                      0.1s
=> [4/6] RUN python -m pip install --upgrade pip      && python -m pip install --no-cache-dir -r requirements.tx 30.3s
=> [5/6] COPY app.py .                                0.2s
=> [6/6] RUN useradd --create-home --shell /usr/sbin/nologin appuser && chown -R appuser:appuser /app 0.6s
=> exporting to image                                  4.2s
=> => exporting layers                                    4.2s
```

- Run the container using created image

docker run -d -p 5000:5000 -e GEMINI_API_KEY=YOUR_GOOGLE_API_KEY -e GEMINI_MODEL=gemini-2.0-flash --name opspilot-local opspilot-ai:local

```
D:\aiops>docker run -d -p 5000:5000 -e GEMINI_API_KEY=AQ.Ab8RN6I4cGI77K87wX9SbH9TJmOYL0h9KK55ULC5rx8qta1Uma -e GEMINI_MODEL=gemini-3.5-flash --name opspilot-local opspilot-ai:local
6fe2378814375f3e540f862ce33030e9779971e4bc8c2e19002e3b7197a9bf40
D:\aiops>
```

curl -f http://localhost:5000/health



- Now remove the docker container
`docker rm -f opspilot-local`

```
D:\aiops>docker rm -f opspilot-local
opspilot-local
D:\aiops>
```

10. PUSH SOURCE TO GITHUB

- Commit and push

| Bash / Git Bash / Linux / macOS | Windows PowerShell | Windows CMD |
|---|---|---|
| git add .
git commit -m "Initial OpsPilot AI v4 setup"
git push -u origin main | git add .
git commit -m "Initial OpsPilot AI v4 setup"
git push -u origin main | git add .
git commit -m "Initial OpsPilot AI v4 setup"
git push -u origin main |

Before push: Configure GitHub Secrets, Variables, and Environments first if you want the first pipeline to complete without reruns.

11. GITHUB ACTIONS PIPELINE AND APPROVALS

- Open Actions tab.
- Confirm AI Agent - Project Health Guardian passes.
- Confirm Build and Deploy validates Python/Docker and pytest.
- Approve docker-publish environment.
- Approve production environment.
- Confirm EC2 deploy succeeds and /health works.

Critical deploy dependency:

docker-publish:
needs: [validate, pytest]

This prevents Docker publish and production deploy when tests fail.

Ⓟ If successful good, pipelines all executes well and good

- Build and Deploy AI Devops Assistant
- AI Agent RCA on CICD

- It wont deploy immediately
- It will ask for permission for deploy

Approval gate

The AI agent has **not changed code** yet.

If you approve the agent to prepare a fix, comment exactly:

```
/ai-agent approve
```



The remediation workflow will then:

1. create a new branch,
2. apply an allow-listed app/CI/CD/Docker/test remediation only,
3. run Python, pytest, Docker build, and container `/health` validation,
4. retry remediation up to the configured limit if validation fails,
5. open a pull request for human review if validation passes.

It will **not** merge the pull request and will **not** deploy production automatically.

Create sub-issue 

 github-actions added ai-agent needs-human-appr... ci-failure 13m ago

 Saibaba-22 1m ago Owner ...

```
/ai-agent approve
```



Add a comment

- Now Type : `/ai-agent approve` and click on comment
- Now check or run
- Build and Deploy AI DevOps Assistant pipeline

✓ Build and Deploy AI DevOps Assistant #6

Summary

All jobs

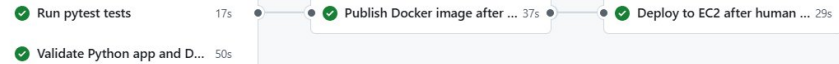
- ✓ Validate Python app and Docker image
- ✓ Run pytest tests
- ✓ Publish Docker image after human approval
- ✓ Deploy to EC2 after human approval

Run details

Manually triggered 2 minutes ago
Saibaba-22 → c8a447c main
Status: Success
Total duration: 2m 10s
Artifacts: 2

deploy.yml

on: workflow_dispatch



• From aws

```
ubuntu@ip-10-5-1-81:~$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
NAMES
dfcl76d74dd4   saibaba22/opspilotai:ee6091606c7eed7bfe7793d0a66b47a26d2b5e67 "gunicorn --bind 0.0..." 7 minutes ago  Up 7 minutes (healthy)  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
p_opspilot
ubuntu@ip-10-5-1-81:~$ docker ps
```

• From azure

```
root@aiops:/home/ordinary# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
NAMES
75083fb63894   saibaba22/opspilotai:762f39be19fdca1acb74404d5703acce43eaa43c "gunicorn --bind 0.0..." 16 seconds ago  Up 13 seconds (healthy)  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
p_opspilot
root@aiops:/home/ordinary#
```

root@ip-10-5-1-81:/home/ubuntu# docker ps

| CONTAINER ID | IMAGE | COMMAND | CREATED | STATUS |
|--------------|---|--------------------------|----------------|-------------------------|
| d9be11dcf99b | saibaba22/opspilotai:c8a447ca3f891a4ef87b40307b2069bc9585b60c | "gunicorn --bind 0.0..." | 14 seconds ago | Up 12 seconds (healthy) |

0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp opspilot
root@ip-10-5-1-81:/home/ubuntu#

- Now Build and Deploy ai assistant done

Update deploy.yml #13

Re-run all jobs

Latest #2

Summary

All jobs

- ✓ Validate Python app and Docker image
- ✓ Run pytest tests
- ✓ Detect available cloud targets
- ✓ Publish Docker image after human approval
- ✓ Deploy approval gate
- ✓ Deploy to AWS EC2
- ✓ Deploy to Azure VM
- ✓ Deployment summary

Re-run triggered 10 minutes ago

Status

Total duration

Artifacts

Saibaba-22

762f39b

main

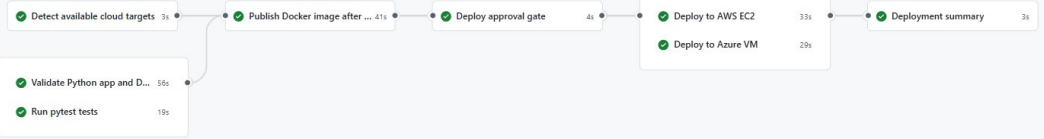
Success

47s

2

deploy.yml

on: push



Actions

New workflow

All workflows

- AI Agent - Human Approved Remediation
- AI Agent - Project Health Guardian
- AI Agent - RCA on CI/CD Failure
- Build and Deploy AI DevOps Assistant

Management

- Caches
- Deployments
- Attestations
- Runners
- Usage metrics
- Performance metrics

All workflows

Showing runs from all workflows

Filter workflow runs



Help us improve GitHub Actions

Tell us how to make GitHub Actions work better for you with three quick questions.

Give feedback



57 workflow runs

Workflow Event Status Branch Actor

AI Agent - Project Health Guardian

AI Agent - Project Health Guardian #14: Manually run by Saibaba-22

now

17s

...

AI Agent - RCA on CI/CD Failure

AI Agent - RCA on CI/CD Failure #17: completed by Saibaba-22

1 minute ago

2s

...

AI Agent - RCA on CI/CD Failure

AI Agent - RCA on CI/CD Failure #16: completed by Saibaba-22

4 minutes ago

29s

...

Update deploy.yml

Build and Deploy AI DevOps Assistant #13: Commit 762f39b pushed by Saibaba-22

7 minutes ago

47s

...

Ⓟ If any file has any errors and pipeline fails.

- Now the pipeline failed and we got error

A. Sample change to check

- For testing we have made a sample change and push or commit in github files directly

<> Code Issues 10 Pull requests Discussions Actions Projects Wiki Security and quality Insights Settings

Build and Deploy AI DevOps Assistant

Update app.py #9

Summary

All jobs

- ✗ Validate Python app and Docker image
- ✗ Run pytest tests
- ✓ Publish Docker image after human approval
- ✓ Deploy to EC2 after human approval

Run details

Triggered via push 1 minute ago

Saibaba-22 pushed 527e87a main

Status

Failure

Total duration

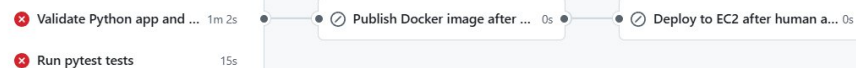
1m 6s

Artifacts

1

deploy.yml

on: push



- Now ai agent RCA on CICD has automatic triggered
- Open the github issues and we can check pipeline failure reason
- Select issue

← AI Agent - RCA on CI/CD Failure

✓ AI Agent - RCA on CI/CD Failure #10

Summary

All jobs

✓ Analyze failed run and create RCA artifact

Run details

Usage

Workflow file

Annotations

1 warning

Analyze failed run and create RCA artifact

succeeded now in 42s

Set up job

Checkout code

Set up Python

Install AI agent dependencies

Resolve failed run ID

Collect workflow metadata and logs

- Click on issues tab

<> Code

Issues 11

Pull requests

Discussions

Actions

Projects

Wiki

Security and quality

Insights

Settings

Issues

Assigned to me

Created by me

Mentioned

Recent activity

Views

Projects

Milestones

All issues

is:issue state:open

Open 11

Closed 0

Author

Labels

Projects

Milestones

Assign

AI RCA: CI/CD failure in run 3063124444

ai-agent

ci-failure

needs-human-approval

#12 · github-actions opened 1m ago

AI RCA: CI/CD failure in run 30628601460

ai-agent

ci-failure

needs-human-approval

#10 · github-actions opened 45m ago

- Select latest issue

AI RCA: CI/CD failure in run 30631244444 #12

 Open


github-actions bot
opened 2m ago

RCA: CI/CD Failure Report

1. Executive Summary

The workflow "Build and Deploy AI DevOps Assistant" (Run ID: 30631244444) failed during the `pytest` stage. The application failed to import the Flask object during test collection due to a variable name mismatch in `app.py`. This prevented the deployment pipeline from proceeding, effectively blocking all downstream Docker publishing and deployment tasks.

2. Failure Evidence

- Workflow: Build and Deploy AI DevOps Assistant
- Failed Job: Run pytest tests
- Failed Step: Run pytest
- Error Detail:

- Scroll down and copy the comment and paste in box and click on comment

Approval gate

The AI agent has **not changed code** yet.

If you approve the agent to prepare a fix, comment exactly:

`/ai-agent approve`



The remediation workflow will then:

1. create a new branch,
2. apply an allow-listed app/CI/CD/Docker/test remediation only,
3. run Python, pytest, Docker build, and container `/health` validation,
4. retry remediation up to the configured limit if validation fails,
5. open a pull request for human review if validation passes.

It will **not** merge the pull request and will **not** deploy production automatically.

Create sub-issue 

 github-actions added `ai-agent` `needs-human-appr...` `ci-failure` 3m ago

Add a comment

Write
Preview

H B I   
  
   

Use Markdown to format your comment

- Copy this comment : `/ai-agent approve`
- Click on comment

Add a comment

Write Preview

/ai-agent approve

Paste, drop, or click to add files

Close with comment

Comment

- Now this AI RCA ckd failures pipeline automatically triggered

Actions

New workflow

All workflows

AI Agent - Human Approved Remediation...

AI Agent - Project Health Guardian

AI Agent - RCA on CI/CD Failure

Build and Deploy AI DevOps Assistant

Management

Caches

All workflows

Showing runs from all workflows

Filter workflow runs

Help us improve GitHub Actions

Tell us how to make GitHub Actions work better for you with three quick questions.

39 workflow runs

Workflow Event Status

AI RCA: CI/CD failure in run 30631244444

AI Agent - Human Approved Remediation #11: Issue comment #12 (comment) created by Saibaba-22

now Queued

AI Agent - Human Approved Remediation

AI RCA: CI/CD failure in run 30631244444 #11

Summary

All jobs

Create AI remediation pull request

Run details

Usage

Workflow file

Triggered via issue 1 minute ago

Saibaba-22 commented on #12 -> 527e87a

Status Success

Total duration 1m 1s

Artifacts 1

ai-agent-remediate.yml

on: issue_comment

Create AI remediation pull r... 48s

- Now check in pull requests tab it is created or not

Issues 11 Pull requests Discussions Actions Projects Wiki Security and quality Insights Settings

Filters is:pr is:open

Labels 13 Milestones 0 New pull request

1 Open 1 Closed

Author Label Projects Milestones Reviews Assignee Sort

AI remediation for CI/CD failure #12

#13 opened 1 minute ago by Saibaba-22 Owner

- Now we got the latest pull request
- Click on the pull request

Ready to merge Code

Conversation 0 Commits 1 Checks 0 Files changed 1 +1 -3





No conflicts with base branch

Merging can be performed automatically.

Merge pull request ▾

You can also merge this with the command line. [View command line instructions.](#)

- Click on merge request.
- Type any thing and click on confirm merge

 **This branch has not been deployed**
 No deployments



Commit message

Merge pull request #13 from Saibaba-22/ai-agent/remediate-issue-12-30631661271

Extended description

AI remediation for CI/CD failure #12

This commit will be authored by 146579091+Saibaba-22@users.noreply.github.com.

Confirm merge
Cancel

- Now one more pipeline triggers and issues get solved.

B. To get issue details

- Now Build and Deploy failed
- AI agent RCA and CICD done
- Now click on issues tab

Issues **10**
Pull requests
Discussions
Actions
Projects
Wiki
Security and quality
Insights
⚙️

ied to me

id by me

oned

t activity

All issues

is:issue state:open

☐ Open **10**
☐ Closed **0**

Author ▾

☐

●

AI RCA: CI/CD failure in run 30628601460
ai-agent
ci-failure
needs-human-approval

#10 · github-actions opened 6m ago

- Now open issue
- We got the RCA id copy it

AI RCA: CI/CD failure in run 30628601460 #10

New issue

Open

github-actions bot opened 7m ago

RCA: CI/CD Failure Report

1. Executive Summary

The workflow "Build and Deploy AI DevOps Assistant" (Run ID: 30628601460) failed on 2026-07-31. The failure originated in the `Run pytest` job. While the Docker image build successfully completed, the test suite failure halted the deployment pipeline, preventing the `docker-publish` and `deploy` stages from executing.

2. Failure Evidence

- Failed Job: `Run pytest`
- Failed Step: `Run pytest`
- Observation: The logs show the workflow reached the `Run pytest` step but terminated in a failed state.

Assignees

No one - [Assign yourself](#)

Labels

`ai-agent` `ci-failure` `needs-human-approval`

Projects

No projects

Milestone

No milestone

- Open now run Human Approved Remediation
- Asks for RCA number, paste it and run

Actions

All workflows

AI Agent - Human Approved Remediat...

AI Agent - Project Health Guardian
AI Agent - RCA on CI/CD Failure
Build and Deploy AI DevOps Assistant

Management

[Caches](#)
[Deployments](#)
[Attestations](#)
[Runners](#)
[Usage metrics](#)
[Performance metrics](#)

AI Agent - Human Approved Remediation

ai-agent-remediate.yml

Filter workflow runs

Help us improve GitHub Actions
Tell us how to make GitHub Actions work better for you with three quick questions.

Give feedback

5 workflow runs

This workflow has a `workflow_dispatch` event trigger.

Run workflow

| Event | Status | Branch | Actor |
|--|---------|--------|------------|
| AI Agent - Human Approved Remediation | Success | main | Saibaba-22 |
| AI RCA: CI/CD failure in run 30626731596 | Failure | main | Saibaba-22 |
| AI Agent - Human Approved Remediation | Success | main | Saibaba-22 |
| AI RCA: CI/CD failure in run 30622346146 | Failure | main | Saibaba-22 |

Use workflow from

Branch: main

RCA issue number to remediate *

Optional failed run ID to fetch logs again

Run workflow

Approval gate

The AI agent has **not changed code** yet.

If you approve the agent to prepare a fix, comment exactly:

`/ai-agent approve`

The remediation workflow will then:

1. create a new branch,
2. apply an allow-listed app/CI/CD/Docker/test remediation only,
3. run Python, pytest, Docker build, and container `/health` validation,
4. retry remediation up to the configured limit if validation fails,
5. open a pull request for human review if validation passes.

It will **not** merge the pull request and will **not** deploy production automatically.

Create sub-issue

github-actions added `ai-agent` `needs-human-approval` `ci-failure` 13m ago

Saibaba-22 1m ago

/ai-agent approve

Open in

Labels

`ai-agent` `needs-hun`

Projects

No projects

Milestone

No milestone

Relationship

None yet

Developer

Create a bra

Notification

You're receiv
subscribed to

Participants

Open in

- Done

C. Pem Key is wrong it shown like this

RCA: CI/CD Pipeline Failure - Run 30626731596

1. Executive Summary

The deployment pipeline for `opapilot-ai` failed during the final stage: "Deploy to EC2 after human approval". While all previous stages (Pytest, Docker validation, and Docker publishing) completed successfully, the deployment job failed at the step `Deploy immutable image tag to AWS EC2`.

2. Failure Evidence

- **Workflow Name:** Build and Deploy AI DevOps Assistant
- **Failed Job:** Deploy to EC2 after human approval (Job ID: 91143924875)
- **Failed Step:** Deploy immutable image tag to AWS EC2
- **Error Context:** The logs show the job started, successfully built the SSH action, but failed during the script execution block. Precise error output from the EC2 shell is truncated, indicating a potential runtime failure in the `appleboy/ssh-action` execution or the deployment script logic.

3. Root Cause

The failure occurred within the `deploy` job. Since the previous `docker-publish` job succeeded, the Docker image is correctly pushed to Docker Hub. The failure is localized to the remote SSH execution on the EC2 instance. Possible causes include:

- **Connectivity/Authentication:** Stale or incorrect `EC2_SSH_KEY` in GitHub Secrets.
- **Environment State:** The EC2 instance disk space, memory, or Docker daemon availability preventing container replacement.
- **Script Failure:** The `set -Euo pipefail` directive in the deploy script triggered an exit due to a command failure (e.g., `docker pull` or `docker run` command not found or not handled by the `appleboy/ssh-action`).

YOU HERE - [Assign yourself](#)

Labels

ai-agent ci-failure needs-human-approval

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Create a branch for this issue or link a pull request.

Notifications

Customize

Subscribe

You're not receiving notifications from this thread.

Participants

No participants

Ops in GitHub Copilot

- My pem key is wrong and need change
- Change the key and run again

Ⓟ We are deployed both clouds

Update deploy.yml #13

Re-run all jobs Latest #2

Summary

All jobs

Validate Python app and Docker image

Run pytest tests

Detect available cloud targets

Publish Docker image after human approval

Deploy approval gate

Deploy to AWS EC2

Deploy to Azure VM

Deployment summary

Re-run triggered 10 minutes ago

Status

Total duration

Artifacts

Saibaba-22 762f39b main Success 47s 2

deploy.yml

on: push

Detect available cloud targets 3s

Publish Docker image after ... 41s

Deploy approval gate 4s

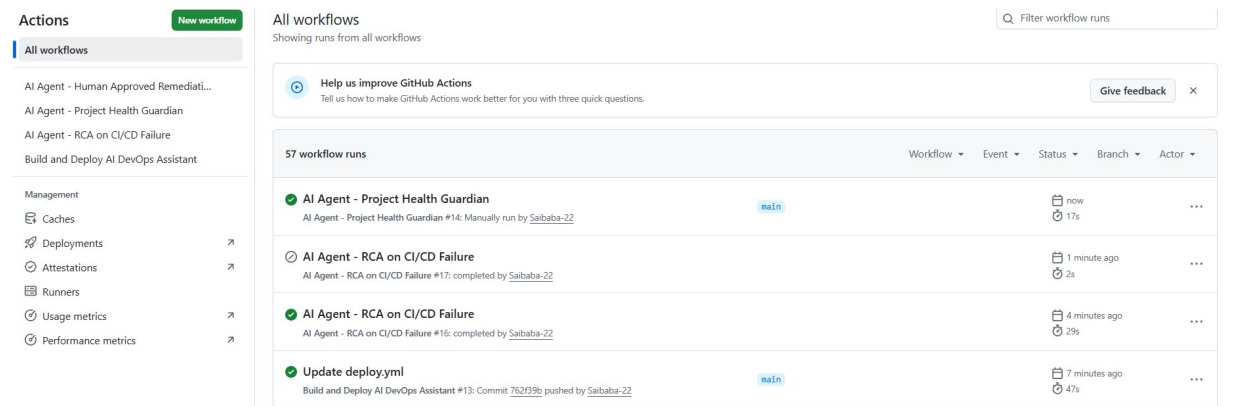
Deploy to AWS EC2 33s

Deploy to Azure VM 29s

Deployment summary 3s

Validate Python app and D... 56s

Run pytest tests 18s



12. ENHANCED AI REMEDIATION V4 ARCHITECTURE

- Agent modes

| Mode | Command | Purpose |
|-----------|--|---|
| analyze | python scripts/ai_ci_agent.py
analyze | Creates RCA artifact/issue content from failed run logs. |
| remediate | python scripts/ai_ci_agent.py
remediate | After approval, attempts fixes with retry loop and validation. |
| doctor | python scripts/ai_ci_agent.py
doctor | Static project health checks for YAML, app, Docker, requirements, tests, ports, policy. |

- Allow-listed remediation only.
- No automatic merge.
- No automatic production deployment.
- PR only after validation passes.
- Validation failure logs feed the next AI attempt.
- Masked placeholder guard prevents `***MASKED***` from being committed.
- Dynamic port discovery prevents blind 5000 assumptions.

13. PROJECT HEALTH GUARDIAN

- Runs on push to main and manual workflow_dispatch.
- Checks YAML syntax, required files, no tracked runtime artifacts, no masked placeholders.
- Checks deploy.yml gates: validate, pytest, docker-publish approval, production approval.
- Checks app.py /health, Dockerfile healthcheck, and discovered port consistency.
- Creates RCA issue if health fails, with /ai-agent approve instructions.

14. AI RCA AND HUMAN-APPROVED REMEDIATION FLOW

- Main pipeline or guardian detects failure.
- AI RCA issue/artifact is created.
- Human reads the RCA and comments /ai-agent approve.

- Remediation workflow checks out code using AI_AGENT_GITHUB_TOKEN.
- AI/deterministic fixes are applied only to allow-listed files.
- Project doctor, Python compile, pytest, Docker build, and /health validation run.
- If validation fails, logs are fed into another attempt.
- If validation passes, AI creates a PR.
- Human reviews and merges PR.
- Main pipeline runs again and reaches normal approvals.

15. CONTROLLED FAILURE TESTS

Lab only: These tests intentionally break main. Use only in a training/lab repo.

Ⓟ Test wrong deploy health port

| Bash / Git Bash / Linux / macOS | Windows PowerShell | Windows CMD |
|--|--|--|
| git checkout main
git pull origin main
edit deploy.yml health check to localhost:50/health
git
add .github/workflows/deploy.yml
git commit -m "Test AI remediation wrong health port"
git push origin main | git checkout main
git pull origin main
edit deploy.yml health check to localhost:50/health
git
add .github/workflows/deploy.yml
git commit -m "Test AI remediation wrong health port"
git push origin main | git checkout main
git pull origin main
REM edit deploy.yml health check to localhost:50/health
git
add .github\workflows\deploy.yml
git commit -m "Test AI remediation wrong health port"
git push origin main |

- Wait for failure/RCA or guardian issue.
- Comment /ai-agent approve.
- Confirm remediation PR restores the discovered app port.
- Confirm no ***MASKED*** placeholders appear.
- Merge PR and verify pipeline passes.

Ⓟ Test intentional pytest failure

| Bash / Git Bash / Linux / macOS | Windows PowerShell | Windows CMD |
|---|---|---|
| git checkout -b training/break-pytest
mkdir -p tests
cat > tests/test_training_failure.py <<'PY'
def test_training_failure():
assert False, "Training pytest failure for AI RCA demo"
PY
git add tests/test_training_failure.py
git commit -m "Training: introduce pytest failure"
git push -u origin training/break-pytest | git checkout -b training/break-pytest
New-Item -ItemType Directory -Force tests Out-Null
@'
def test_training_failure():
assert False, "Training pytest failure for AI RCA demo"
'@ Set-Content tests
est_training_failure.py
git add tests
est_training_failure.py
git commit -m "Training: introduce pytest failure"
git push -u origin training/break-pytest | git checkout -b training/break-pytest
mkdir tests 2>nul
(
echo def test_training_failure^(^):
echo assert False, "Training pytest failure for AI RCA demo"
)> tests est_training_failure.py
git add tests
est_training_failure.py
git commit -m "Training: introduce pytest failure"
git push -u origin training/break-pytest |

16. TROUBLESHOOTING MATRIX

⑨ Common failures and exact meaning

| Symptom | Likely cause | Fix / next action |
|--|---|---|
| RCA is generic and says AI provider failed | AI_AGENT_MODEL unavailable or API key issue | Set AI_AGENT_MODEL=gemini-2.0-flash and ensure AI_AGENT_API_KEY is valid. |
| Checkout fails: could not read Username for github.com | AI_AGENT_GITHUB_TOKEN secret missing/invalid | Create PAT with repo + workflow permission and save as secret. |
| GitHub rejects workflow file push | Token lacks workflow permission | Use AI_AGENT_GITHUB_TOKEN in checkout and GH_TOKEN for PR creation. |
| Invalid workflow YAML after remediation | AI wrote masked placeholders or bad YAML | v4 masked guard should block this. Restore expressions and rerun. |
| pytest fails in remediation | Fix did not solve app/test issue | Review validation artifact; v4 retries up to AI_AGENT_MAX_ATTEMPTS. |
| Docker publish login fails | Docker secret/token issue | Manual fix: update DOCKERHUB_USERNAME/DOCKERHUB_TOKEN. |
| EC2 SSH fails | EC2 key/host/security group issue | Manual fix: update EC2_HOST, EC2_SSH_KEY, SG port 22. |
| App Internal Server Error | Gemini model/key issue or app runtime exception | Check docker logs; set GEMINI_MODEL=gemini-2.0-flash; verify key format. |

17. WHAT AI CAN AND CANNOT FIX

⑨ Boundary of automation

| Can fix after approval | Cannot directly fix; AI should report manual action |
|---|--|
| app.py syntax, route, /health, runtime code mistakes | Expired/invalid Docker Hub token |
| Dockerfile command/port/healthcheck issues | Wrong/missing EC2 SSH private key |
| deploy.yml health checks, Docker mappings, YAML/policy issues | AWS security group blocked or EC2 stopped |
| requirements and pytest issues | Gemini quota/account/API key value problems |
| masked placeholder corruption | Cloud provider outage or Docker Hub outage |
| tracked runtime artifacts or missing ignore rules | Any secret value that must be changed outside the repository |

18. FINAL CHECKLIST AND SUBMISSION EVIDENCE

- New repo created and source pushed.

- GitHub secrets and variables configured.
- docker-publish and production environments configured.
- Local pytest passed.
- Project doctor Status: PASS.
- Local Docker build/run passed.
- Guardian workflow passed.
- Main pipeline passed.
- Docker publish and production approvals tested.
- At least one controlled failure demo performed.
- AI RCA issue reviewed.
- /ai-agent approve remediation PR created.
- AI PR reviewed and merged.
- Final pipeline passed after remediation.

⑩ Recommended evidence

| Item | Details |
|------------------------------|--|
| GitHub repo URL | Your new OpsPilot AI repository. |
| Successful Actions run URL | Shows guardian + main pipeline success. |
| Docker Hub image/tag | Published image. |
| EC2 app URL/screenshot | App and /health response. |
| RCA issue screenshot | AI RCA generated after controlled failure. |
| AI remediation PR screenshot | PR created after /ai-agent approve. |
| Doctor output | Status: PASS artifact or screenshot. |

END OF OPSPILOT AI V4 FRESH SETUP GUIDE

Errors

```
Q (venv) D:\aiops>python app.py
Traceback (most recent call last):
  File "D:\aiops\app.py", line 2, in <module>
    from flask import Flask, request, render_template_string
ModuleNotFoundError: No module named 'flask'
```

A install requirements.txt by
Pip install -r requirements.txt

Now python app.py